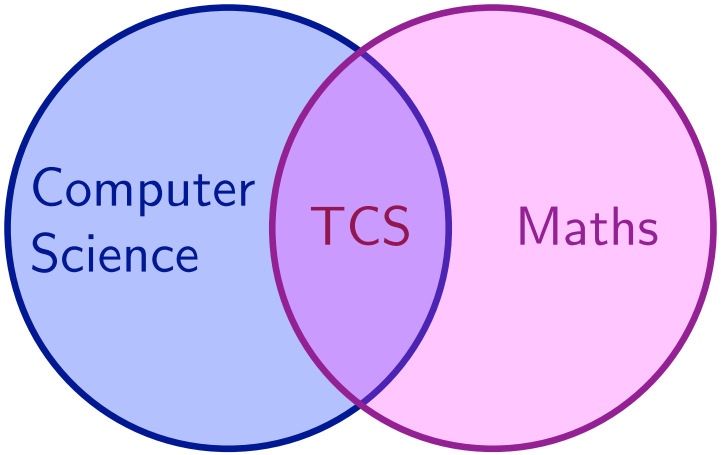
Breaking the Vigenère cipher

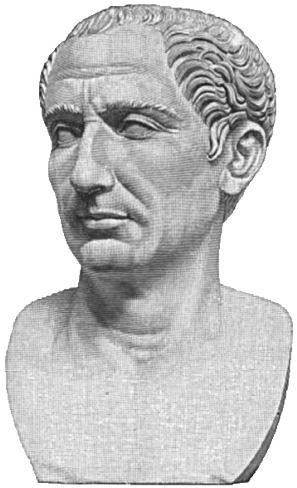
Dillon Mayhew

Professor of Theoretical Computer Science at
the School of Computer Science, University of Leeds.



UNIVERSITY OF LEEDS

Caesar cipher



The Caesar cipher shifts every letter three places:

$$\begin{array}{ccccccccc} A & B & C & D & \cdots & W & X & Y & Z \\ \downarrow & \downarrow & \downarrow & \downarrow & \cdots & \downarrow & \downarrow & \downarrow & \downarrow \\ D & E & F & G & \cdots & Z & A & B & C \end{array}$$

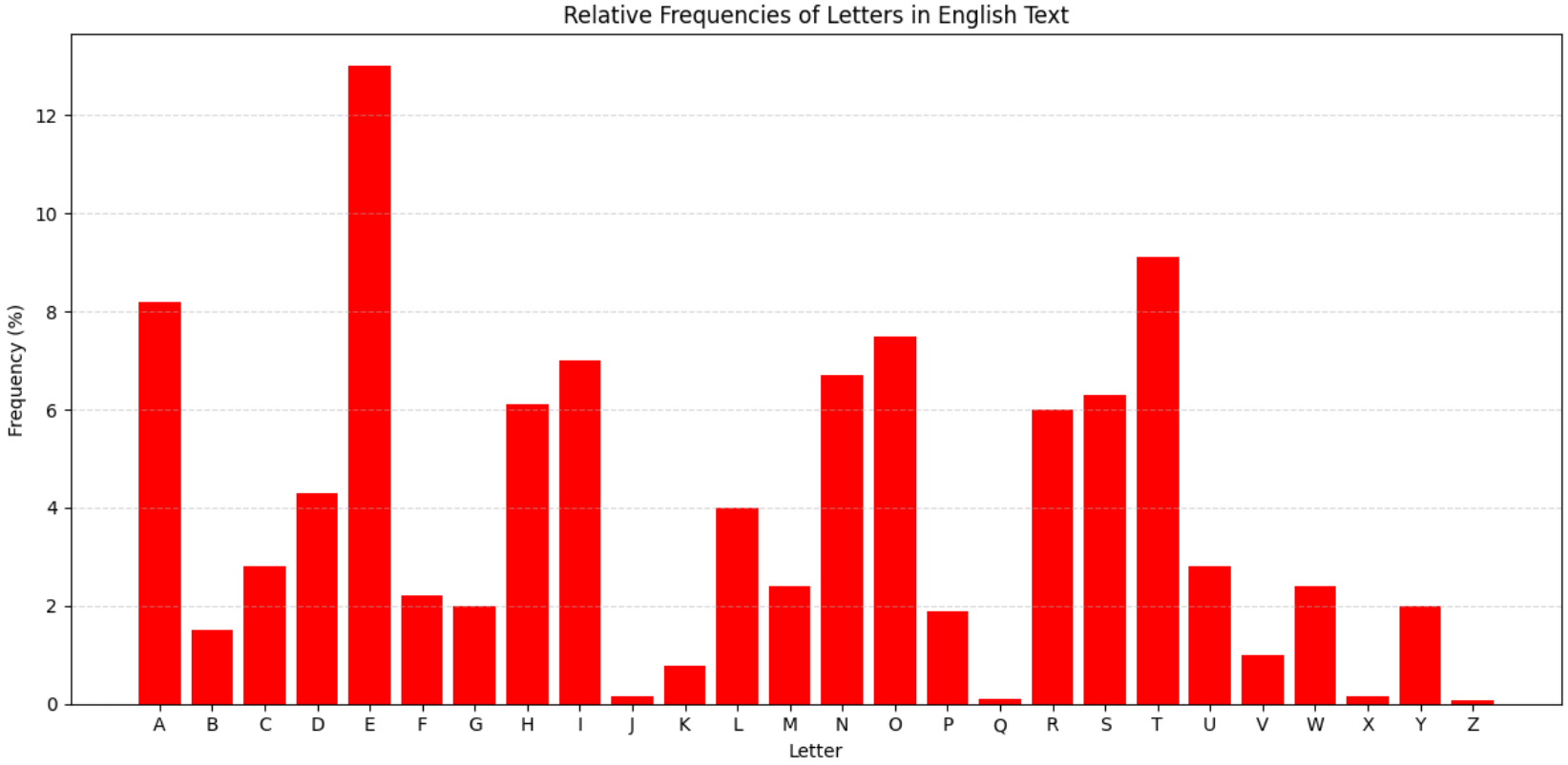
WAKE UP CALL is encrypted as ZDNH XS FDOO.

LQ D SLFNOH is decrypted as IN A PINCH.

In general, a shift cipher shifts each letter the same number of places.

Shift ciphers

Shift ciphers are easy to break because the letters in English follow a distinctive pattern of frequencies.



We can try out different possible shifts until these frequencies match the frequencies in the encrypted text.

Shift ciphers

Can we break this shift cipher?

QBYH C ZCHX GSMYFZ CH NCGYM IZ NLIOVFY
GINBYL GULS WIGYM NI GY
MJYUECHA QILXM IZ QCMXIG FYN CN VY
UHX CH GS BIOL IZ XULEHYMM MBY CM MNUHXCHA
LCABN CH ZLIHN IZ GY
MJYUECHA QILXM IZ QCMXIG FYN CN VY
FYN CN VY FYN CN VY
FYN CN VY FYN CN VY
QBCMJYL QILXM IZ QCMXIGFYN CN VY

Index of coincidence

Frequencies of letters in English are very far from a distribution that is chosen uniformly at random.

We can measure how close a distribution is to uniform by calculating the index of coincidence.

This measures the probability that when we randomly select a pair of letters from a text of N letters, the pair will be identical.

$$\text{Index of coincidence} = \frac{\#A(\#A-1)}{N(N-1)} + \cdots + \frac{\#Z(\#Z-1)}{N(N-1)}$$

Index of coincidence

If each letter in the text is chosen uniformly at random, the index of coincidence will be close to

$$\frac{1}{26} \approx 0.0385.$$

English text has a index of coincidence that is closer to 0.0700.

Any text encrypted with a shift cipher will also have a index of coincidence close to 0.0700.

Vigenère cipher

The Vigenère cipher was regarded as being unbreakable between 1550 and 1850 (approximately). To encode, we follow these steps:

- Choose a key word. E.g. ALERT
- Repeat the key below the input text.
- Shift each letter by the number of places corresponding to that letter of the key.

Input:	*A*	*T*	*T*	*A*	*C*	*K*		*A*	*T*		*D*	*A*	*W*	*N*
Key:	*A*	*L*	*E*	*R*	*T*	*A*		*L*	*E*		*R*	*T*	*A*	*L*
Shift:	0	11	4	17	19	0		11	4		17	19	0	11
Output:	*A*	*E*	*X*	*R*	*V*	*K*		*L*	*X*		*U*	*T*	*W*	*Y*

Breaking the Vigenère cipher

This text has been encoded using the Vigenère cipher. Can we break it?

LYMAK APJAV ZCTRZ FQPSZ WAEIH RENOE VEQSK GTEBC KGILZ RKRPS WBMSC KPRXM GXQAH
SGVAG EEACM QUXDI QRUSU CQSGM INIFC FDRKB PWIIQ PVSGK PNVLP UAJCP AFERF ELZTQ DIPJB
UGVTZ ANESE PWSVL TUZNY MCXQG YEHNZ VVXTE QAVLT SWBMF DRKBP WIIQP VSGJA JCPAF
EGJIX IEUQP TGZRA AXKPB WXRYM JKRVK PRPXW VGPCR CFBZQ ZTRVQ VLTPA YGIEU CEKRV
KPRUM MDWAV LHFNG JIXIL VIIHK QBPMI YMAVL DLOUV EAFBN DSJKB UGESM MAVYG VABHX
WVRHP KAVIA FMCKC EPWJT KRGHT UEVVL PTWYQ VTUXR PGXCQ AFVPN QAIQN MMEAJ XIAGF
VPNQA IMINI FNMZV BUKWX JPBYI SDGZC WIVZC KIRVB BVLTX ZBYRJ GANPH PJSRF XWVUV
HQNUZ NYMCX NEKKW KMAGH IYMZV LTPAN KHLYG JQYAU IUCXQ VAPCV NDGQT ELZVT YEHEW
GQJPY IGKXL RANDS PTWAU XGZKG QVSZO RUXXE ONPIA VXUCR IZBUG RSIMJ VLTZV FKHTF NGJIQ
FIPQR HKZVE XDIAB VLPKB UGKGF EAWTH TWHNH JELRT WIRVQ VLTPI YYENJ ZRSYX IMQOS
GVMKR PPEIG KSC

Breaking the Vigenère Cipher

If the key word has length P, then every Pth letter is encrypted using the same shift cipher.

Extracting every Pth letter should produce a distribution that is close to the English alphabet.

We can detect this by calculating the index of coincidence.

Breaking the Vigenère Cipher

Once we have found the key length P, we need to find the key word.

The 0th, Pth, $2P$th letters are all shifted by the same number. We can guess this shift by comparing frequencies.

The 1st, $(P+1)$th, $(2P+1)$th letters are shifted by the same number, so we repeat this procedure. We then do the same with the 2nd, $(P+2)$th, $(2P+2)$th letters, and so on.

In this way, we can guess the key word.

Thanks for listening!

Code used in this demonstration can be found here:

`https://github.com/dillon128/Vigenere`

You can implement SageMath code online at:

`https://sagecell.sagemath.org/`